

May 12, 2026

1 T1: DC Analysis, DC Sweep, Parameter Sweep

1.1 Hand Calculation

1. Calculate the signal transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_2}{R_1 + R_2}$$

2. Since $R_1 = R_2$:

$$V_{\text{out}} = 0.5V_{\text{in}}$$

V_{in} (V)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
V_{out} (V)	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50

V_{in} (V)	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
V_{out} (V)	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1.00

V_{in} (V)	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
V_{out} (V)	1.05	1.10	1.15	1.20	1.25	1.30	1.35	1.40	1.45	1.50

V_{in} (V)	3.1	3.2	3.3
V_{out} (V)	1.55	1.60	1.65

3. Parameter sweep:

$$V_{\text{out}} = 3.3 \text{ V} \cdot \frac{R_2}{1 \text{ k}\Omega + R_2}$$

R_2 (k Ω)	1	2	3	4	5	6	7	8	9	10
V_{out} (V)	1.65	2.20	2.475	2.64	2.75	2.8286	2.8875	2.9333	2.97	3.0

1.2 Cadence Verification

(b)

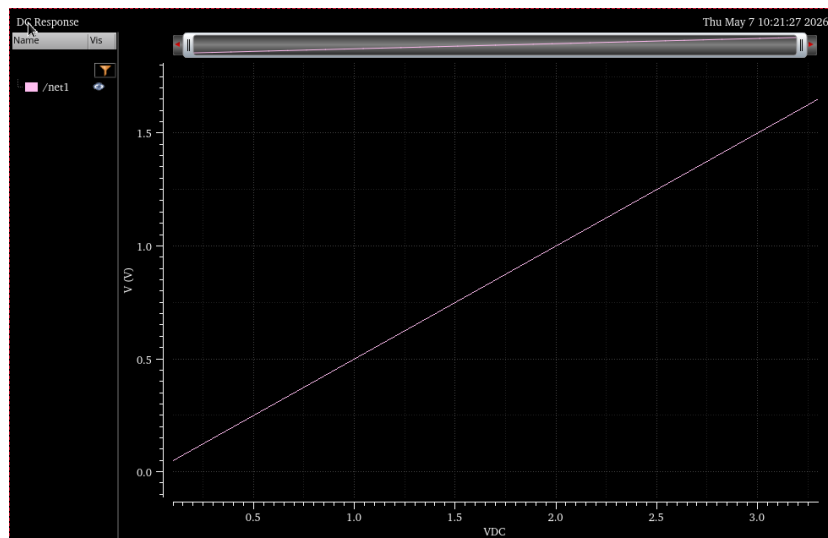


Figure 1: DC Sweep Simulation

(c)

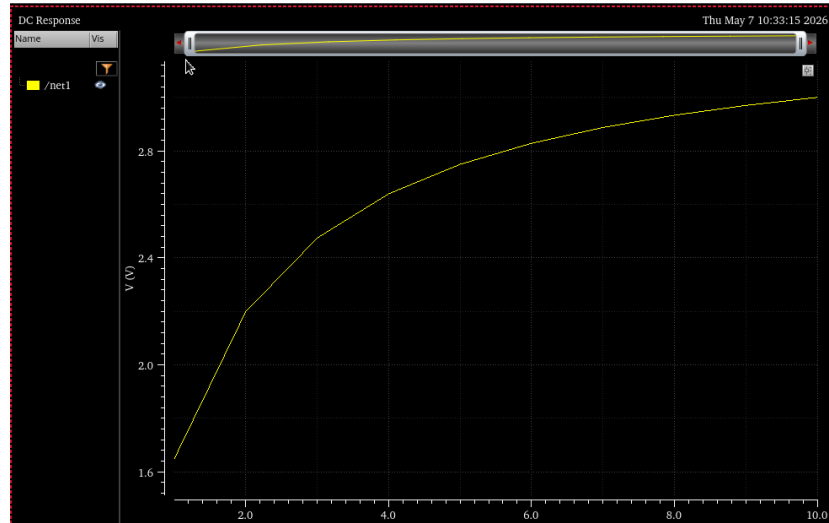


Figure 2: Parameter Sweep Simulation

2 T2: Transient Analysis and AC Analysis

1. Transfer function:

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC} = \frac{1}{1 + j\omega RC}$$

Magnitude response:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + (2\pi f RC)^2}}$$

Phase response:

$$\angle H(j\omega) = -\arctan(\omega RC)$$

2. -3 dB cut-off frequency:

The cut-off frequency is defined as the frequency where the magnitude drops to

$$\frac{1}{\sqrt{2}}$$

of the DC gain.

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}}$$

$$\omega_c RC = 1$$

$$\omega_c = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

Given:

$$R = 12.5 \text{ k}\Omega$$

$$C = 8 \text{ pF}$$

$$RC = 12.5 \times 10^3 \times 8 \times 10^{-12} = 10^{-7} \text{ s}$$

Therefore:

$$f_c = \frac{1}{2\pi \times 10^{-7}} \approx 1.59 \text{ MHz}$$

3. AC and transient simulation results:

$$f_c \approx 1.588 \text{ MHz}$$

Phase shift $\approx -45^\circ$

AC Analysis

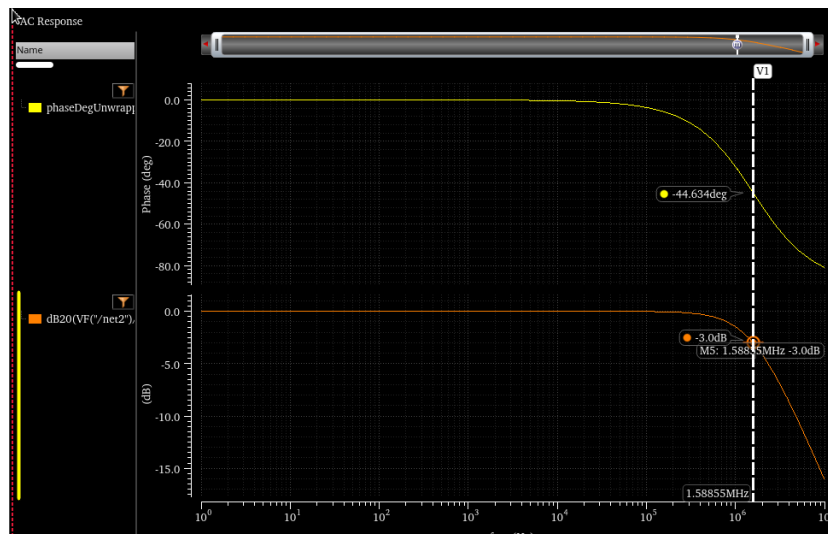


Figure 3: AC Response

Transient Analysis

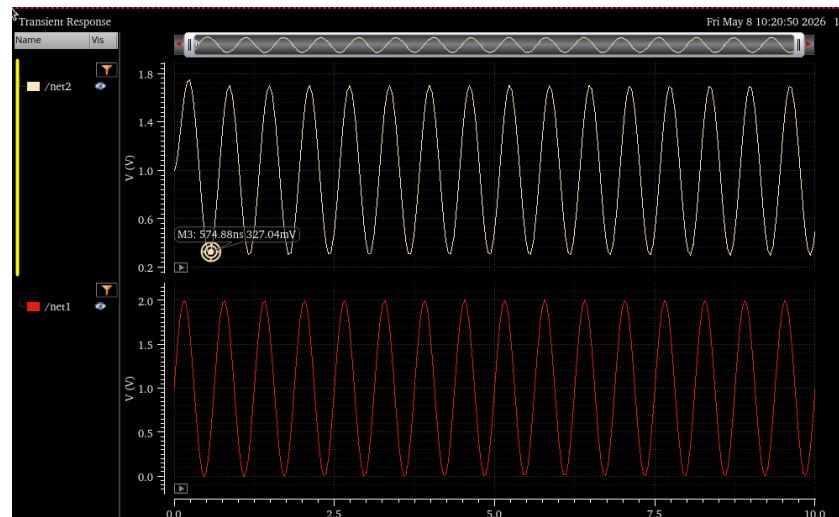


Figure 4: Transient Response

3 T3: Transistor Transfer Characteristic Curve

1. Simulated I_D versus V_{DS} characteristic:

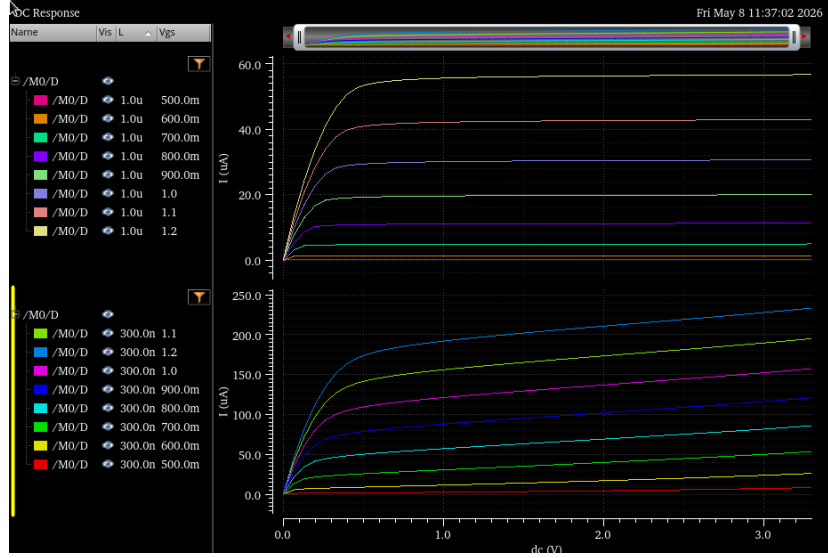


Figure 5: I_D - V_{DS} Characteristic

2. Channel-length modulation parameter:

$$\lambda = \frac{I_{D2} - I_{D1}}{I_{D1}V_{DS2} - I_{D2}V_{DS1}}$$

3. Discussion:

- For short-channel devices ($L = 0.35\mu\text{m}$), λ is larger because channel-length modulation is stronger.
- As V_{GS} increases, λ also increases slightly.